SHED MEASUREMENT NOTE

Emergency Buffers Over Time

A SHED Repeated Cross-Section Measurement Note, 2015–2025

Arne Kotte

AUTHOR
Arne Kotte

DOCUMENT TYPE
Measurement note

DATE
May 2026

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Abstract

This paper revises the multi-year SHED emergency-buffer draft after benchmarking its constructed series against the Federal Reserve's published $400 cash/equivalent trend. The revision changes the central interpretation. A pre-2020 component-built cash/equivalent measure should not be treated as the same object as the Federal Reserve's official “would cover a $400 emergency expense completely using cash or its equivalent” series. The official published series is 54, 56, 59, 61, 63, 64, 68, 63, 63, 63, and 63 percent for 2015–2025, with 2021—not 2019—as the high point. The author's component-built diagnostic series is materially higher in 2015–2019 and aligns much more closely from 2020 onward, when the public-use files include `pay_casheqv`. The paper therefore reframes its contribution as a measurement note rather than a continuous trend claim. The reliable empirical core is narrower: 2020–2025 direct cash/equivalent comparisons and 2021–2025 covariate-adjusted repeated cross-sectional associations. In those windows, adults without $400 cash/equivalent capacity report far worse financial outcomes, and adults lacking both $400 cash/equivalent capacity and three-month savings remain much more likely to be financially vulnerable and under bill pressure after controls. These are descriptive associations, not causal estimates.

1. Introduction

The prior draft attempted to build a harmonized 2015–2025 SHED emergency-buffer series from public-use microdata. An external-style review identified a serious problem: the pre-2020 constructed cash/equivalent series did not match the Federal Reserve's published $400 cash/equivalent trend. That critique is correct.

The correction is substantive, not cosmetic. The original draft reported a 2019 high of 72.0 percent in harmonized cash/equivalent $400 capacity. The Federal Reserve's published SHED trend instead reports 63 percent in 2019 and a high of 68 percent in 2021. Treating the author's pre-2020 constructed component measure as the same object as the official cash/equivalent series overstated pre-2020 emergency liquidity and changed the narrative.

This revised paper does three things.

1. It benchmarks the constructed/direct microdata diagnostic against the official Federal Reserve published series. 2. It separates official trend evidence from component-built diagnostic evidence. 3. It limits the main respondent-level analysis to windows where the public-use variables support the claim more cleanly: 2020–2025 for direct cash/equivalent comparisons and 2021–2025 for richer adjusted associations.

The result is a more modest but more reliable paper: a measurement note documenting what the SHED public-use files can support, where harmonization fails, and what evidence remains useful after correcting the central trend problem.

2. Data and terminology

The data are Federal Reserve SHED public-use repeated cross-sections. The paper no longer describes the dataset as a panel. The public-use files include annual adult respondent samples, and while some SHED releases include panel-related weights or identifiers for specific linked respondents, this draft does not estimate within-person transitions or person fixed effects.

The script downloads or reuses annual public-use CSV ZIP files, extracts selected variables, and writes harmonized diagnostic tables. After correcting the 2013 URL pattern to the official Federal Reserve file name, it finds public-use files for 2013–2025. The current measurement note begins in 2015 because the core emergency-response variables needed for this analysis appear from 2015 onward in the inspected files.

The key distinction is between two objects:

- **Official cash/equivalent $400 trend:** the Federal Reserve published report series for adults who would cover a $400 emergency expense completely using cash or its equivalent.
- **Constructed/direct diagnostic:** the author's microdata variable. Before 2020 it is constructed from component response options such as checking/savings/cash and credit-card-paid-in-full. From 2020 onward it uses or closely tracks `pay_casheqv`.

These are not interchangeable. The paper now treats the official series as the benchmark and the constructed/direct series as a diagnostic.

3. Benchmarking the central measure

Table 1 shows the corrected benchmark. The pre-2020 constructed diagnostic is 8 to 13 percentage points above the Federal Reserve's published cash/equivalent series. From 2020 onward, the gap is much smaller.

Table 1. Official benchmark and constructed/direct diagnostic.

Year	Official cash/equiv $400	Constructed/direct diagnostic	Difference (pp)
2015	54.0	66.0	12.0
2016	56.0	68.7	12.7
2017	59.0	71.6	12.6
2018	61.0	69.1	8.1
2019	63.0	72.0	9.0
2020	64.0	64.6	0.6
2021	68.0	67.7	-0.3
2022	63.0	65.0	2.0
2023	63.0	63.3	0.3
2024	63.0	62.7	-0.3
2025	63.0	63.1	0.1

The correct trend claim is therefore the Federal Reserve's: emergency cash/equivalent capacity rises from 54 percent in 2015 to 68 percent in 2021, then returns to roughly 63 percent in 2022–2025. The earlier “2019 high” claim is withdrawn.

This benchmark also clarifies the paper's empirical boundary. Pre-2020 component construction is useful for diagnostics and code exploration, but it is not validated as a replacement for the official published measure. Any future version should reproduce the Federal Reserve series exactly before extending it.

4. Other annual measures

Some emergency-response measures remain useful as repeated cross-sectional descriptive series. Table 2 reports selected annual rates. These are still subject to question-wording and harmonization checks, but they are less central than the official cash/equivalent benchmark.

Table 2. Selected annual weighted rates from inspected public-use files.

Year	Official cash/equiv $400	Card over time	Unable $400 now	Three-month savings	Doing okay	Financially vulnerable
2015	54.0	17.4	13.9	48.1	68.6	31.4
2016	56.0	20.0	12.1	48.7	70.0	30.0
2017	59.0	17.9	12.0	50.5	73.7	26.3
2018	61.0	15.6	12.4	51.6	74.9	25.1
2019	63.0	15.8	12.6	53.3	75.4	24.6
2020	64.0	15.6	12.6	55.4	75.1	24.9
2021	68.0	13.9	10.8	59.1	77.7	22.3
2022	63.0	15.8	13.1	54.1	73.1	26.9
2023	63.0	15.6	12.8	53.8	72.2	27.8
2024	63.0	15.2	13.0	55.0	72.9	27.1
2025	63.0	15.3	12.2	54.8	72.9	27.1

The broad pattern is now more restrained. Official $400 cash/equivalent capacity improves through 2021 and then falls back to about 63 percent. Three-month savings follows a similar broad shape, rising to 59.1 percent in 2021 and settling near 55 percent in 2024–2025. Financial vulnerability is lowest in 2021 and higher in 2022–2025.

The paper no longer displays bill pressure as a full annual trend. The review was right that a late-period bill-pressure column invited comparison across non-comparable definitions. Bill pressure remains useful in 2020–2025 and 2021–2025 analyses where the relevant outcome is clearly denominated, but it should not carry a full-period trend claim.

5. Direct 2020–2025 cash/equivalent comparisons

The cleanest respondent-level cash/equivalent window begins in 2020, when `pay_casheqv` appears directly in the public-use files. Table 3 pools 2020–2025 and compares adults by cash/equivalent $400 capacity.

Table 3. 2020–2025 direct cash/equivalent cross-tabs.

Group	Outcome	N	Weighted rate (%)	95% CI (%)
No cash/equiv $400	Doing okay	23,432	44.2	43.5–44.9
Has cash/equiv $400	Doing okay	48,386	90.4	90.1–90.7
No cash/equiv $400	Financially vulnerable	23,432	55.8	55.1–56.5
Has cash/equiv $400	Financially vulnerable	48,386	9.6	9.3–9.9
No cash/equiv $400	Bill pressure	12,640	41.6	40.6–42.5
Has cash/equiv $400	Bill pressure	23,989	7.2	6.8–7.6

The inability-to-pay-$400 outcome is omitted from this main table because it is mechanically related to the $400 response item. It remains useful for internal diagnostics, but it is not an independent hardship outcome when the right-hand-side variable is cash/equivalent $400 capacity.

The non-mechanical comparisons remain large. Adults without cash/equivalent capacity are much less likely to report doing okay and much more likely to be financially vulnerable. In the years where bill-pressure outcomes can be used, the gap is also large.

6. Adjusted 2021–2025 associations

The richest covariate coverage begins in 2021. Table 4 reports approximate weighted linear-probability models for 2021–2025. The predictor is low buffer: lacking both cash/equivalent $400 capacity and three-month savings. Controls include income, age, education, race/ethnicity, housing tenure, employment, children, region, and survey wave fixed effects.

Table 4. 2021–2025 adjusted low-buffer associations.

Outcome	N	Low-buffer coefficient (pp)	SE (pp)	95% CI (pp)
Financially vulnerable	60,168	37.7	0.5	36.7–38.8
Bill pressure	36,627	25.3	0.6	24.0–26.5

The mechanical inability-to-pay-$400 model is excluded from the main table. It is not a valid independent outcome when low-buffer status includes the $400 emergency-response measure.

The remaining adjusted associations are still descriptive, but they are informative. Low-buffer adults are much more likely to report financial vulnerability and bill pressure after observed covariates and wave fixed effects are included. This does not prove that buffers cause lower hardship, but it shows that the low-buffer measure is not simply a restatement of income, tenure, or age in the public-use files.

7. Measurement implications

The central lesson is methodological. SHED is useful for emergency-buffer research, but harmonization must be benchmarked against official published series before trend claims are made. Component-built variables can look plausible and still fail an official-topline check.

A sound revision path is now clear:

1. reproduce the Federal Reserve's official $400 cash/equivalent trend exactly; 2. create a year-by-year codebook map of question wording, universes, response options, and missing-data treatment; 3. compare component-built and direct measures in overlap years; 4. keep mechanical outcomes out of main adjusted models; 5. move bill-pressure outcomes into definition-consistent windows; 6. add a portable replication package with file hashes and codebook provenance.

This paper implements the first, fourth, and fifth corrections partially. It does not yet complete the full variable map or a publication-ready replication package.

8. Limitations

First, the analysis is descriptive and repeated cross-sectional. It does not follow the same adults over time.

Second, pre-2020 component-built cash/equivalent capacity is not validated as equivalent to the Federal Reserve's official measure. The official series should be used for the main trend.

Third, survey timing should be read as survey wave timing, not calendar-year outcomes. Many questions describe current conditions or experiences over the prior 12 months.

Fourth, uncertainty estimates are approximate and do not implement full survey-design variance estimation.

Fifth, the replication package remains local and preliminary. A public package should include source URLs, hashes, codebook versions, software versions, and one-command reproduction.

9. Conclusion

External review materially improved the paper. The original central trend claim was not reliable because the pre-2020 constructed cash/equivalent measure did not match the Federal Reserve's official series. The corrected paper withdraws the 2019-high claim and reframes the work as a SHED measurement note.

The corrected empirical message is narrower. The official $400 cash/equivalent series rises from 54 percent in 2015 to 68 percent in 2021, then returns to about 63 percent in 2022–2025. In the cleaner 2020–2025 direct-measure window, adults without cash/equivalent capacity report much worse financial outcomes. In 2021–2025 adjusted models, adults lacking both $400 cash/equivalent capacity and three-month savings are substantially more likely to be financially vulnerable and under bill pressure.

That is useful descriptive evidence. The best next step is a replication-grade harmonization appendix and package that can reproduce official SHED toplines before extending them.

Data and replication note

The replication workflow is script-based and rebuilds the harmonized panel from public SHED files.

Main outputs:

- `data/processed/shed_2015_2025_harmonized_panel.csv`
- `tables/shed_2015_2025_harmonized_trends.csv`
- `tables/shed_official_cash_equiv_benchmark.csv`
- `tables/shed_2020_2025_cash_equiv_crosstabs.csv`
- `tables/shed_2021_2025_adjusted_lpm.csv`
- `tables/shed_harmonized_multi_year_tables.md`

A publication-ready replication package should use repository-relative paths, include file hashes, cite annual SHED public-use files and codebooks, and provide one-command reproduction.

References

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